LUNA ELLIOTT

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Publications may appear under the name Luke Elliott.

RESEARCH PROFILE

I work in topological algebra, geometric group theory and semigroup theory. Author of 20 research papers (11 published), including publications in Transactions of the American Mathematical Society, Advances in Mathematics, Journal of Algebra and Mathematics of Computation.

SELECTED PAPERS (10 OF 20)

See the complete list by clicking here.

- Automatic continuity, unique Polish topologies, and Zariski topologies on monoids and clones. L Elliott, J Jonušas, Z Mesyan, J Mitchell, M Morayne, Y Peresse. Transactions of the American Mathematical Society 376 (11), 8023-8093 2023
- Sufficient conditions for a group of homeomorphisms of the Cantor set to be two-generated. C Bleak, L Elliott, J Hyde. Journal of the Institute of Mathematics of Jussieu 23 (6), 2825-2858 2024
- A short proof of Rubin's theorem. J Belk, L Elliott, F Matucci. Israel Journal of Mathematics 267 (1), 157-169 2025
- Polish topologies on endomorphism monoids of relational structures. L Elliott, J Jonušas, JD Mitchell, Y Péresse, M Pinsker. Advances in Mathematics 431, 109214 2023
- E-disjunctive inverse semigroups. L Elliott, A Levine, J Mitchell. Journal of Algebra. 2025
- Topological embeddings into transformation monoids. S Bardyla, L Elliott, J Mitchell, Y Péresse. Forum Mathematicum 36 (6), 1537-1554 2024
- A description of Aut(dVn) and Out(dVn) using transducers. L Elliott. Groups, Geometry & Dynamics 17 (1), 2023
- The Diophantine problem in Thompson's group F. L Elliott, A Levine Mathematics of Computation. 2025
- Counting monogenic monoids and inverse monoids. L Elliott, A Levine, JD Mitchell. Communications in Algebra 51 (11), 4654-4661 2023
- The Zariski Topology on Homeomorphism groups. L Elliott. Topology and its Applications, 2026

DEGREES

PhD, University of St Andrews (2021)
Thesis: On constructing topology from algebra

ACADEMIC SERVICE

Co-organiser, Manchester Semigroup Seminar. Referee for peer-reviewed mathematics journals.

Document last updated 31/08/2026

EMPLOYMENT HISTORY

- 2024-present: University of Manchester Heilbronn Fellow. The Heilbronn Institute for Mathematical Research (HIMR), is a UK national research institute funded by GCHQ. My time is divided equally between working on fundamental mathematical research projects relating to GCHQ's long-term strategic interests (which are classified), and my own personal research (which I can publish). Fellowships are awarded competitively based on a personal research proposal, and equate to a personal grant of £200K over 3 years.
- 2022-2024: Visiting Assistant Professor, Dept. Mathematics, Binghamton University, New York State University
- 2022: Research Fellow, University of St Andrews, Programming (in GAP) algorithms for working with the GNS rational groups, automorphisms of the shift and Thompson groups.
- 2018-2021: PhD Student at the University of St Andrews
- 2017: Laidlaw Undergraduate Research and Leadership Program: Algorithms for commutative semigroups (awarded £4000)

TEACHING

See more teaching information by clicking here.

- 2022-2024: Lecturer for six undergraduate mathematics courses (300+ lecture hours). Responsible for course delivery, lecture preparation, office hours, assessment design, writing and marking examinations, quizzes and coursework.
- 2018-2021: Tutor, University of St Andrews. Delivered tutorials for undergraduate mathematics courses during my PhD.

SELECTED TALKS (5 OF 26)

See the complete list by clicking here.

- Reconstructing Topologies on Monoids: Invited speaker at ATTI (Online, University of Udine, Italy) 2026
- Thompson Monoids: Invited speaker at RIMS conference (Kyoto, Japan) 2026
- Thompson Partition Monoids and Jónsson-Tarski algebras: Invited speaker at TCA (Guimarães, Portugal) 2026
- Topologizing Endomorphism Monoids: AAA (Vienna, Austria) 2026
- Semigroup Theory: Algebra seminar (Binghamton NY, USA) 2024

RESEARCH FUNDING

Awarded three competitive Heilbronn Institute research grants (principal applicant once; co-applicant twice).

INTERESTS

Maths, Programming, Indoor Climbing, and Conversational Japanese.

REFERENCES

Available on request.